

FERNADO GEORGE ANAK MANI

Kuching, Sarawak | +60 14-320 7322 | milobio351@gmail.com | github.com/Fernado03
linkedin.com/in/fernado-george | www.fernado03.my

PROFESSIONAL SUMMARY

Final-year Bachelor of Computer Science (Data Science) student at Universiti Malaysia Sabah, currently based in Kuching, Sarawak, with a CGPA of 3.75/4.00 and practical experience developing and deploying AI/ML and computer-vision systems. Scheduled to complete an AI/ML Development internship through 28 August 2026; available for full-time employment from September 2026. Expected graduation: 30 November 2026.

PROFESSIONAL EXPERIENCE

Breakfast Byte Sdn. Bhd. | AI/ML Development Intern

16 March 2026 - Present

Kuching, Sarawak

- Developed KI Konnekt, a Gradio-based RAG assistant for a German client using Ollama and OpenAI workflows for document analysis, web search, structured outputs, and visual content generation.
- Engineered hybrid BM25/FAISS retrieval and multi-format document-processing pipelines with streaming responses, persistence, usage tracking, and downloadable artifacts.
- Built a Python computer-vision pipeline that converted car-wash CCTV footage into structured vehicle-visit records using session detection, license-plate OCR, vehicle classification, and wash-action recognition.
- Containerized and deployed the pipeline using Docker, Amazon ECR, Amazon ECS tasks, and Amazon S3 for automated processing, storage, and traceable daily reporting.

TECHNICAL SKILLS

- **Programming:** Python, SQL, R, Java, C++, JavaScript, HTML, CSS
- **Machine Learning and Data:** Data analysis, data preprocessing, feature engineering, machine learning, deep learning, model training and evaluation, computer vision, NLP, RAG, Large Language Models
- **Libraries and AI Platforms:** PyTorch, TensorFlow, OpenAI API, Gemini, Ollama, FAISS, BM25, Gradio
- **Web and Development:** React, Vite, Node.js, Tailwind CSS, Git, VS Code
- **Databases:** MongoDB, SQLite, relational SQL databases
- **Cloud and Deployment:** AWS ECS, Amazon ECR, Amazon S3, Docker, Google Cloud Platform

EDUCATION

Universiti Malaysia Sabah | Bachelor of Computer Science (Data Science) with Honours 2022 - Present

- Current CGPA: 3.75/4.00; Dean's List
- Expected graduation: 30 November 2026

Sarawak Matriculation College | Science Module II

2021 - 2022

- CGPA: 4.00/4.00; Best Computer Science Student Award

SELECTED PROJECTS AND ACHIEVEMENTS

Multimodal Emotion Recognition — Final Year Project | Best Research Award 2026

- Conducted a comparative study of contextual and statistical fusion strategies for robust multimodal emotion recognition using speech, visual, and text features.
- Developed and evaluated models using WavLM, DINOv2, ModernBERT, Bi-GRU, and ensemble-learning methods under Gaussian noise and missing-modality conditions.
- Achieved a 71.6% F1-score using an ensemble strategy, while the Contextual Bi-GRU architecture provided the strongest balance between robustness and computational efficiency.

KinaVis - AI Medical Scribe | Gold Medal - Festival Idea Sabah 2025

- Architected an AI-powered medical documentation system using Google Speech-to-Text, Gemini 2.5 Pro, and Retrieval-Augmented Generation.
- Implemented vector retrieval using all-MiniLM-L6-v2 to generate contextually grounded clinical records and reduce hallucinations.
- Reduced documentation time by approximately three to four hours per day and administrative costs by 30% during prototype evaluation.

MatchKami - AI School-NGO Matching Platform | 2nd Prize - JomHack Varsity Challenge 2025

- Developed an AI-powered matching system connecting under-resourced schools with suitable NGOs and donors.
- Implemented smart need-matching, a verified donor network, and an accessible AI chatbot on Telegram using Python and natural-language processing.
- Designed the solution to improve access to educational resources for underserved and rural communities.

LEADERSHIP AND CO-CURRICULAR INVOLVEMENT

PMFKI Student Association | Head of Academic and Career Affairs (2023–2024)

- Supported association administration, event coordination, communications, and academic and career-development activities for students.

Hackathon Team Lead and Lead Developer | 4+ Hackathons

- Led teams in more than four hackathons, including MarineHack and JomHack, while managing intensive 48-hour development sprints.
- Oversaw solution architecture, task delegation, development, team coordination, and final product pitching.
- Contributed to teams that secured one Gold Medal and one 2nd Prize.

REFERENCES

Dr. James Mountstephens

Senior Lecturer and Final-Year Project Supervisor
Faculty of Computing and Informatics, Universiti Malaysia Sabah
jmountstephens@ums.edu.my | +60 12-834 9044

Charissa Phua

Co-Founder, Director and Industrial Supervisor
Breakfast Byte Sdn. Bhd.
charissa.phua@breakfast-byte.com | +60 11-6744 1298